

Robustness of ML-based temperature downscaling

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1. Why downscale a climate model?

Running climate simulations at very high resolutions is slow and compute expensive



Train a statistical/ML model to learn patterns on pairs of LR and HR weather snapshots

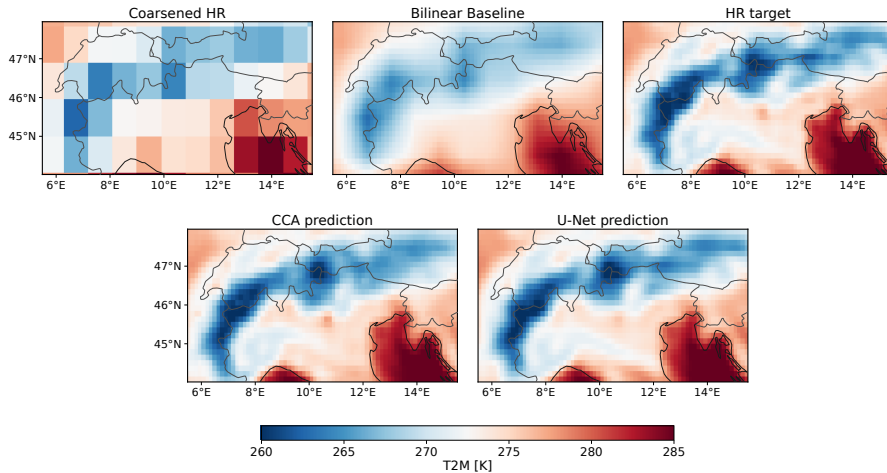


Run LR simulations and retrieve fine-scale sub-grid information that the simulation doesn't capture via the model (called "downscaling")

2. Context

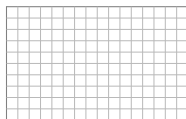
- Canonical Correlation Analysis (CCA), applied after reducing the fields to an Empirical-Orthogonal-Function (EOF) basis, is a well-established traditional statistical method for downscaling
- ML as a replacement for traditional statistical models:
 - ① Does it recover scientifically relevant fine-scale information beyond what CCA can?
 - ② Does it generalize across climates better than CCA?
- Cross-climate generalization has received limited direct testing.

3. Visualizing a sample task



4. Global 2-m temperature fields as points in a vector space

Global T2M field



2624 × 1280 grid cells

flatten
→

Same field, vectorized

$$\mathbf{T} = \begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ \vdots \\ T_{3,358,720} \end{pmatrix}$$

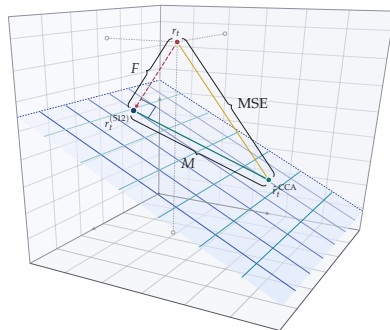
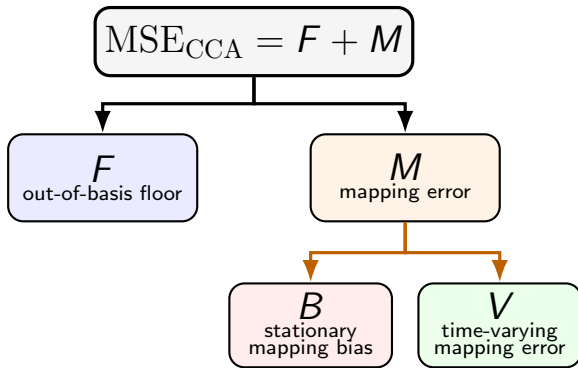
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Geometric interpretation

$$\mathbf{T} \in \mathbb{R}^{3,358,720}$$

Each daily global field is one point in this high-dimensional space.

5. Exact CCA decomposition in the frozen target space



F: residual structure the 512-EOF target space (the slanted blue plane in the illustration) cannot represent.

M: error made while predicting residual structure that is representable.

Schematic: the labelled segments represent the corresponding error components for one day. F and M are the averages over all evaluation days of their squared area-weighted magnitudes.